

Employee Onboarding Companion

AI-Powered Assistant for New Employee Onboarding

Team 6 • AMPCUS Hackathon

React

Next.js

Node.js

Python

OpenAI GPT

Swagger

Project Overview

The Employee Onboarding Companion is an AI-powered web application that assists new employees throughout the onboarding journey.



Simplify Onboarding

Single point of access for all onboarding information



24/7 AI Assistance

GPT-powered assistant available any time, any device



Reduce HR Workload

Automate repetitive Q&A — free HR teams for strategic work



Better Experience

Personalised, instant responses improve employee satisfaction

Problem Statement

1 New employees struggle to find scattered onboarding information

2 HR teams repeatedly answer the same common questions

3 Onboarding content lives across disconnected systems

4 Process is slow, inconsistent, and frustrating for everyone

Goal: One intelligent AI assistant for all onboarding needs.

The Challenge

Proposed Solution

An intelligent onboarding chatbot powered by OpenAI GPT



Understands Questions

Natural language understanding handles varied employee queries



GPT-Powered Responses

OpenAI GPT delivers accurate, context-aware answers



Centralised Knowledge

Single assistant for HR policies, IT setup, and benefits

Key Benefits



Faster onboarding



Less HR dependency



Better engagement



Scalable & always on

System Architecture



Project Structure

Frontend

- ✓ React.js — Component-based UI
- ✓ Next.js — SSR & routing
- ✓ Node.js — Runtime environment

OpenAI GPT API

Backend

- ✓ Python (FastAPI) — REST APIs
- ✓ API Layer — Business logic
- ✓ LLM Integration — Prompt engineering

Swagger / OpenAPI

Technology Stack

Frontend

React.js

Framework

Next.js

Runtime

Node.js

Backend

Python (FastAPI)

AI Model

OpenAI GPT API

API Docs

Swagger / OpenAPI

Frontend Architecture



Chat UI

Responsive chat interface for employee Q&A sessions



Input Handling

Form controls, input validation, keyboard shortcuts



Display Responses

Formatted AI reply rendering with markdown support



API Calls

Fetch calls from React to the Python backend REST layer



SSR (Next.js)

Server-side rendering for fast initial page loads

Flow

User



React UI



Next.js



Backend API

Backend Architecture

1 Receive and validate frontend HTTP requests

2 Process and sanitise user input

3 Build contextual prompts for the LLM

4 Call the OpenAI GPT API

5 Return structured, formatted AI responses

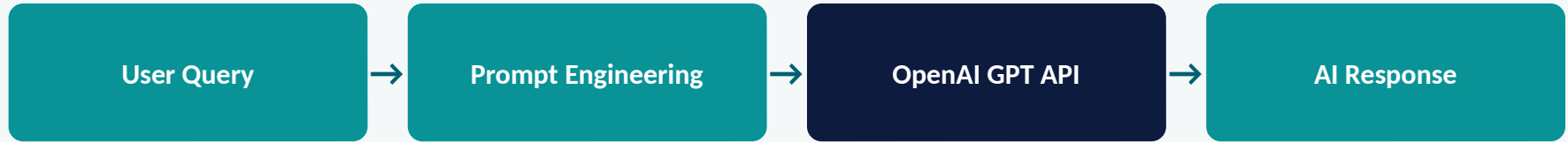
Components

REST APIs

Business Logic

LLM Integration

LLM Integration — OpenAI GPT



HR Questions

Benefits, leave policy, payroll, onboarding steps



IT Setup Guidance

Laptop setup, VPN, account provisioning, software



Company Policies

Code of conduct, dress code, remote work guidelines



General Onboarding

First-day schedules, introductions, team structure

API Communication Flow

Step 1

Employee

Types query in chat

Step 2

Frontend — React + Next.js

Sends HTTP POST to backend

Step 3

Backend — Python FastAPI

Validates and builds GPT prompt

Step 4

OpenAI GPT API

Returns AI-generated response

Step 5

Backend processes response

Formats and validates output

Step 6

Frontend displays result

Renders in chat UI for employee

Swagger / OpenAPI Documentation



Interactive Testing

Try API endpoints directly from the browser — no Postman needed



Clear API Contracts

Auto-generated request/response schemas for every endpoint



Developer Onboarding

New devs understand the API in minutes, not hours

Endpoints

POST

/chat

Send a chat message

POST

/ask

Ask an onboarding Q

GET

/health

Health check probe

Key Features



AI-Powered Assistant

GPT backbone handles natural language queries accurately



Chat-Based Interface

Familiar messaging UX — no learning curve for employees



Instant GPT Responses

Sub-second AI answers with context-aware prompt design



Responsive React UI

Works on desktop, tablet, and mobile out of the box



Modular Backend

Clean separation of concerns makes features easy to add



Swagger API Documentation

Interactive docs for all backend endpoints

Future Enhancements

1

RAG Knowledge Base

Connect company docs, wikis, and HR portals for grounded answers

2

Voice Assistant

Enable voice queries for hands-free onboarding support

3

Slack / Teams Bot

Embed the assistant directly in collaboration platforms

4

Multilingual Support

Support global workforces with multi-language responses

5

Personalised Paths

Adaptive onboarding flows based on role, department, and location

6

HR Analytics Dashboard

Insights on common questions and onboarding progress metrics

Employee Onboarding Companion

Built by Team 6 • AMPCUS Hackathon

- Faster, smoother onboarding for every new hire
- Reduced HR workload through AI self-service
- Smarter, always-on employee experience
- Scalable AI-based foundation for the future

Thank You • Questions & Discussion